Siddharth Sondhi

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Education

- **San Jose State University** — San Jose, CA
  Master of Science in Computer Science; GPA: 3.81 — *Aug. 2024 – May 2026*
- **UC Riverside** — Riverside, CA
  Bachelor of Science in Computer Science; GPA: 3.65 — *Sep. 2020 – Jun. 2024*

Projects

- **OpenGL Graphics Pipeline Renderer**: Real time renderer built in C++ using OpenGL and GLSL shaders. Implemented a 3D renderer featuring Blinn-Phong lighting, forward and deferred rendering, directional shadow mapping with PCF for soft shadows, model loading via Assimp, instanced rendering, cubemap skyboxes, framebuffer based post processing effects, and normal mapping. Designed a custom scene management system as well, supporting rendering of multiple scenes by selection through GUI integrated with Dear ImGui.
- **N-Body Galaxy Simulation with Octree**: Gravitational particle simulation in C++ using Euler integration. Optimized from $O(n^2)$ brute force to O(n log n) via a Barnes-Hut octree algorithm, enabling real time simulation of many particles. Created with OpenFrameworks.
- **Ray Tracer**: CPU ray tracer with reflections, refractions, and shadows. Implements Phong shading and multi bounce recursive rays to produce realistic images. Built in C++.
- **Ray Marcher**: Rendered 3D shapes defined by signed distance functions using a ray marching algorithm. Implemented in C++ and OpenFrameworks.
- **AI-Assisted VR Anatomy Education App**: Unity-based virtual reality environment integrating local LLM inference, speech recognition, and function calling pipelines for natural language interaction. Conducted a formal comparative user study between AI-assisted and traditional VR interaction methods. Developed in C# with Unity as part of MSCS thesis work.

Experience

- **Verve Group**
  Software Engineering Intern — *Summer 2023*
  - **Role**: Contributed to a data science project extracting and analyzing device and location data to derive user insights for targeted advertising. Investigated correlations between demographic data (e.g., US Census income brackets) and click behavior.
  - **Languages & Tools**: Python, Pandas, Git, JupyterLab
- **InMobi**
  Software Engineering Intern — *Summer 2021*
  - **Role**: Automated a previously manual data upload process by building a new tab in a web application. Implemented file parsing, error checking, and user feedback in JavaScript and HTML. Used Python to make API calls and upload parsed data to the company database.
  - **Languages & Tools**: JavaScript, Python/Django, Git, HTML/CSS, Docker

Skills

- **Languages**: C++, GLSL, Python, C#, Java
- **Graphics & Rendering**: OpenGL, GLSL Shaders, Real-Time Rendering, Rasterization, Graphics Pipeline, Shader Programming, Phong Lighting, Ray Tracing, Ray Marching, Signed Distance Functions (SDFs), Framebuffer & Post-Processing, Cubemaps, Instanced Rendering, Geometry Shaders, Shadow Mapping
- **Simulation & Math**: N-Body Simulation, Particle Systems, Barnes-Hut Algorithm, Octree Spatial Partitioning, Euler Integration, Linear Algebra, 3D Math, Coordinate Systems, Matrix Operations
- **Tools & Frameworks**: Git/GitHub, OpenFrameworks, Assimp, Dear ImGui, Unity, CMake, Command Line